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7 Write a program to implement Logistic Regression (LR) algorithm in python

CSV FILE

Age,EstimatedSalary,Purchased

22,20000,0

35,45000,0

26,43000,0

27,58000,0

32,100000,1

25,90000,1

40,108000,1

42,80000,1

39,50000,0

45,22000,0

37,78000,1

52,150000,1

48,134000,1

33,31000,0

50,88000,1

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class LogisticRegressionScratch:
```

```

def __init__(self, learning_rate=0.1, iterations=1000):
    self.lr = learning_rate
    self.iterations = iterations
    self.weights = None
    self.bias = None

def _sigmoid(self, z):
    return 1 / (1 + np.exp(-z))

def fit(self, X, y):
    num_samples, num_features = X.shape
    self.weights = np.zeros(num_features)
    self.bias = 0

    for _ in range(self.iterations):
        linear_model = np.dot(X, self.weights) + self.bias
        y_predicted = self._sigmoid(linear_model)

        dw = (1 / num_samples) * np.dot(X.T, (y_predicted - y))
        db = (1 / num_samples) * np.sum(y_predicted - y)

        self.weights -= self.lr * dw
        self.bias -= self.lr * db

def predict(self, X):
    linear_model = np.dot(X, self.weights) + self.bias
    y_predicted = self._sigmoid(linear_model)
    return np.array([1 if p >= 0.5 else 0 for p in y_predicted])

```

```
df = pd.read_csv("logistic_regression_data.csv")
```

```
X = df[["Age", "EstimatedSalary"]].values
```

```
y = df["Purchased"].values
```

```
X_mean = np.mean(X, axis=0)
```

```
X_std = np.std(X, axis=0)
```

```
X_scaled = (X - X_mean) / X_std
```

```
model = LogisticRegressionScratch(learning_rate=0.1, iterations=1000)
```

```
model.fit(X_scaled, y)
```

```
predictions = model.predict(X_scaled)
```

```
accuracy = np.mean(predictions == y) * 100
```

```
print(f"Learned Weights: {model.weights}")
```

```
print(f"Learned Bias: {model.bias:.4f}")
```

```
print(f"Model Accuracy: {accuracy:.2f}%")
```

```
sample = np.array([[38, 90000]])
```

```
sample_scaled = (sample - X_mean) / X_std
```

```
pred = model.predict(sample_scaled)
```

```
print(f"\nPrediction for sample [Age: 38, Salary: $90,000]: Class {pred[0]}")
```

OUTPUT

